A manual for installing FASTCORE, rFASTCORMICS_v2, COBRA Toolbox V3, GUROBI & MATLAB R2024b for MacOS

MacOS (tested for silicon macs, Sequoia)

1- Installing MATLAB R2024b

1. Go to https://nl.mathworks.com/ and create an account using your student/academic email.
2. Sign-in to the same website using your account.
3. Open https://nl.mathworks.com/downloads/ then click on "Show more" on the left side.
4. Choose R2024b version then download MATLAB installer.
5. Check these MATLAB toolboxes during installing:

Bioinformatics Toolbox (required)
Curve Fitting Toolbox (required)
Optimization Toolbox
Parallel Computing Toolbox
Signal Processing Toolbox
SimBiology
Statistics and Machine Learning Toolbox
Symbolic Math Toolbox

2- Installing GUROBI Optimizer :

2.1. Downloading GUROBI

2.1.1. Go to the GUROBI Download Center: https://www.gurobi.com/downloads/

2.1.2. Register and log in

2.1.3. Get a free license here: https://www.gurobi.com/academia/academic-program-and-licenses/
Save this license key, you will need it later.

2.1.4. Download the GUROBI Optimizer (version 12.0.3)
Execute the gurobi installer file downloaded by clicking on it (gurobi12.0.3_macos_universal2.pkg), follow the installation instructions.
Accept the default path `/Library/gurobi`<ver>/.

2.1.5. Enter the gurobi key from your license into the command line. Should look like this:
`grbgetkey _some number_`

2.1.6. Execute this file: `/Library/gurobi1203/macos_universal2/bin/gurobi_cl`
should get the output: `[Process completed]`

3- Install the latest git for MacOS: https://git-scm.com/install/mac

Use **homebrew** to install git from the command line

Check if **homebrew** is already installed:

```
brew -help.
```

If the output is: `command not found` install brew using following command, else continue to the **install git** section below.

```
xcode-select -install
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Install homebrew to: /opt/homebrew

```
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >>
~/.zprofile
eval "$(/opt/homebrew/bin/brew shellenv)"
```

verify installation

```
brew doctor
brew -version
```

install git: using following command in the terminal

```
brew install git
```

check that git is successfully installed:

```
git -help
```

if it outputs

```
usage: git …
```

then git is installed and you can move on

4- Installing COBRA Toolbox V3:

3.1. Run MATLAB as administrator

3.2. Double check that the setfield command has only this path "/Applications/MATLAB_R2024b.app/toolbox/matlab/datatypes/setfield.m" by running:

```
> which -all setfield
```

If there are other paths, remove them such as these examples:

```
> rmpath('C:\Program Files\MATLAB\R2019b\toolbox\finance\ftseries\')
> rmpath('C:\Program Files\MATLAB\R2019b\toolbox\sl3d\sl3d\')
```

3.3. In your command window, type the following lines (one by one):

```
> cd C:\gurobi1200\win64\matlab
> gurobi_setup
> savepath
```

3.4. Download COBRA Toolbox from https://github.com/opencobra/cobratoolbox by opening Git Bash, and then running

> git clone --depth=1 https://github.com/opencobra/cobratoolbox.git cobratoolbox

3.4. In MATLAB, add COBRA Toolbox folder to MATLAB paths by:

Clicking on Set Path
Clicking on Add with subfolders
Browsing to find your COBRA Toolbox folder, and then clicking on Select Folder
Clicking on Save

3.5. In MATLAB, navigate through your "cobratoolbox" folder.
Then run the following line in your command window:

```
> initCobraToolbox.m
```

You should see this line:

```
gurobi        active        1        1      1      1      -      -      -
```

3.6. To select GUROBI as your solver, run this line in your command window:

```
> changeCobraSolver('gurobi')
```

Note: If MATLAB crashes and closes during installation, open MATLAB and run again the `initCobraToolbox.m` and you might want to double-check that cobratoolbox is still added to the path using which `-all initcobratoolbox`. If the command is not found, repeat step 3.4.

5- Installing FASTCORE and rFASTCORMICS_v2:

5.1. From Git Bash:
> git clone https://github.com/sysbiolux/rFASTCORMICS
Or manually:

from « Code », click on “Download ZIP”

5.2. From MATLAB command window:

```
addpath(genpath('/PATH OF THE DOWNLOADED rFASTCORMICS/'))
```

5.3. A version of FASTCORE already exists in the COBRA Toolbox. However, we will use an updated version from the rFASTCORMICS github. Delete the installation of FASTCORE in the cobratoolbox folder:

- execute: open initCobraToolbox.m
- check the path on top of the editor window, and navigate into the folder where the cobratoolbox is placed on your computer (with command line, finder/browser, or the current folder tab in matlab)
- within the cobratoolbox folder, navigate to the following directory: ./src/dataIntegration/transcriptomics/
- delete the FASTCORE folder

5.4. If you are able to call the following functions (and more related functions) in the command window of MATLAB then the installation is done. To check the location of the files execute the following commands and check that the file paths match

- `open fastcc` *(from the rFastcormics_v2/scripts/fastcore folder)*
- `open fastcore` *(from the rFastcormics_v2/scripts/fastcore folder)*
- `open rFastcormics4cobra_v2` *(from the rFastcormics_v2/scripts/rfastcormics folder)*

Debugging the installation

Problem: Failed installation with crashing MATLAB

Solution: you may need to remove *cobratoolbox* and *cplex* paths, and add them again using

```
% Restoring all default paths, please note this would remove paths for other non-default
MATLAB tools
restoredefaultpath
% Adding the CPLEX path
addpath('C:\Program Files\IBM\ILOG\CPLEX_Studio1210\cplex\matlab\x64_win64')
setenv('ILOG_CPLEX_PATH','C:\Program
Files\IBM\ILOG\CPLEX_Studio1210\cplex\matlab\x64_win64')
Cplex % should return Logical 1
% Checking if there is only this path for setfield command:
% "/Applications/MATLAB_R2019b.app/toolbox/matlab/datatypes/setfield.m "
which -all setfield
% Saving permanently the new paths
savepath
```

Problem: MATLAB is still not detecting *Cplex* command

1.1 Go to Edit the System Environment Variables (Quickly: enter 'env' in the windows search bar)
Click on Environment Variables...
1.2 In the second box named System variables, check for variables named CPLEX DIR or CPLEX BINARY, click on it and remove it.
Click on New...
1.3 Add the following variable:
Name: ILOG_CPLEX_PATH
Value: /actual/path/to/your/cplex/installation
1.4 Click ok to save the variable.
2.1 Open MATLAB R2019b or below
In the Home/Environment tab, click on Set Path
2.2 Verify that NO path to IBM CPLEX is in the list, else remove it.
At that step, you can also double check that cobratoolbox and rFASTCORMICS folder are properly added.
Click on save to close the Set Path panel.
2.3 In the MATLAB terminal, first load the GLPK solver with changeCobraSolver('glpk'), ignore the warnings.
You should receive the logical 1 as a result.
2.4 Add your actual IBM path to MATLAB with the following code:
addpath('/actual/path/to/your/cplex/installation') (note: genpath is not necessary here and might also be the reason of causing path conflicts)
2.5 Now change the solver to ibm_cplex by using changeCobraSolver('ibm_cplex')
This should give you a logical 1 as answer and you are ready to go.
2.6 Happy modeling.
The steps of 2.1 - 2.3 need to be repeated in each new MATLAB session (from the command window if you are using live script) if you plan to use the ibm_cplex solver.

Problem: Eval errors right after MATLAB startup

1.1 Log in with an admin account.
1.2 Go into C:/Program Files/MATLAB/R2019b/toolbox/local folder and locate the file 'pathdef.m'
1.3 Copy this file into the C:/Users/Public/Documents folder so that local accounts have access to the file.
1.4 Log out the admin account.
1.5 Log in local account
1.6 Local should go to C:/Users/Public/Documents folder a copy the 'pathdef.m' file into their documents MATLAB folder (C:/Users/USERNAME/Documents/MATLAB
1.7 Student can now run MATLAB application without the eval errors.

Note: Paths added with *addpath* or *genpath* are usually added to this pathdef.m file which cannot be guaranteed for this copied pathdef.m file.

Problem: COBRA toolbox is not detecting git during installation

1.1 Go to Edit the System Environment Variables (Quickly: enter 'env' in the windows search bar)

1.2 Click on Environment Variables...
1.3 In the first box named System variables, open “PATH”, then add these 2 paths:

c:\Program Files\Git\bin\git.exe
c:\Program Files\Git\cmd

1.4 Click ok to save the variable.

Problem: Version compatibility (MATLAB is crashing due to version compatibility with CPLEX)
Solution: Other MATLAB guaranteed comptability versions **can be also tested:**
CPLEX 12.10 + MATLAB 2019b (or older till 2015b 8)
CPLEX 12.9 + MATLAB 2018b (or older till 2015b 8)
CPLEX 12.8 + MATLAB 2017b 9.3 (or older till 2015b 8)

Problem: COBRAToolbox folder is detected to be owned by Git
Error message must contain:

- fatal: **detected dubious ownership** in repository at 'C:/Users/cobratoolbox/cobratoolbox-master'

Solution: Run the following code in Git bash

- git config --global --add safe.directory C:/Users/cobratoolbox/cobratoolbox